$$\langle \hat{q}_i(\tau)\hat{q}_j \rangle_{\hat{\rho}_\mathcal{E}} = \langle \hat{q}_i(\tau) \rangle_{\hat{\rho}_\mathcal{E}} \langle \hat{q}_j \rangle_{\hat{\rho}_\mathcal{E}} = 0, \tag{5.7}$$